

Operating System (OS) Final Exam

Exam time 2 hours (PM 1:30- 3:30)

1. Assuming the current memory configuration is shown in the figure, the white area represents the unused portion. Please specify the first and most inappropriate cases when the usage is First Fit, Best Fit, Next Fit, and Worst Fit,

(a) Which area will the new itinerary be assigned to?

(b) What is the size of the internal debris produced?

Answer :

(a) First Fit : B

Next Fit : C

Best Fit : D

Worst Fit : B

(b) First Fit : 500K

Next Fit : 12K

Best Fit : 2K

Worst Fit : 500K

2. Suppose a system has a logical address length of 32 and a page length of 64K. May I ask:

(a) In logical addresses, how many bits are occupied by page numbers and offset values?

(b) In which range should the page number value fall in?

Answer :

(a) 16 (b) 0 ~ 65535

3. Please explain the requirements paging mechanism. Because the operating system is passive, your answer must also explain how the operating system knows the program's requirements.

Answer:

In a requirement pagination system, all page breaks of processes are stored in auxiliary memory. During execution, if a page to be accessed is not in main memory, hardware will generate a pagination error interrupt notification to the operating system, which then takes responsibility Copy the page to main memory.

4. Please explain the **Advantages** and **Disadvantages** of using larger pages versus smaller pages.

Answer:

Advantages of using smaller pages: In the case of internal fragmentation, each process wastes half a page on average, so using smaller pages reduces memory waste. From a regional perspective, using smaller pages can more accurately represent the regionality of the process.

Disadvantages of using smaller paginations: the pagination table is larger. Pagination errors are more likely to occur.

Advantages of using larger pagination: fewer pages, making the page table smaller. This not only saves space but also reduces the time spent searching page breaks. Larger pages can cover more commands and data, reducing the chance of pagination errors. From an I/O perspective, to load the same data, using larger pages only requires fewer pages, which takes less time than loading multiple smaller pages.

Disadvantages of using larger tabs: From the perspective of internal fragmentation, using larger pages results in a significant waste of memory space. Larger pages also struggle to show regional characteristics.

5. In a system with 32-bit addresses, a two-level pagination method is used. A logical address is divided into three parts: the first 9 bits are the index for the outer page table, the middle 11 bits are for the inner page table, and the rest are page

offsets. What is the size of each page? How large is the system's address space?

Answer :

Outer index	Inner index	Page offset
9	11	12

Because the page offset is 12 bits, the size of a page is $2^{12} = 4 \text{ K}$.

The address space is $2^9 * 2^{11} * 4 \text{ K} = 4 \text{ GB}$.

6.Please explain how you can break the deadlock? What factors need to be considered?

Answer:

There are two ways to break the deadlock: the first is to directly terminate some processes, making the conditions for loop waiting void; Another method is to recover some resources from the deadlock process to other processes, so that the no-first-priority condition does not apply. Factors to consider include how to choose your itinerary, retrospectives, and hunger phenomena.

7.Please explain how to prevent deadlocks. Which of these conditions is unavoidable?

Answer:

For a deadlock to occur, four conditions must be met simultaneously, so any one of the conditions to prevent a deadlock can be prevented.

Mutual exclusion conditions are unavoidable.

8.What does this mean?(a) OS Critical Section -Method 1 (b) OS Critical Section -Method 2 (c) OS Critical Section -Method 3 How does each algorithm operate?

Answer : Omitted

9.How can deadlocks be prevented?

Answer : Omitted

10. Please run the following P0 to P4:

Deadlock Avoidance: Banker's Algorithm

EN / 中文

Next Step

Reset

Process	Allocation	Max	Need
P0	0, 1, 0	3, 5, 3	7, 4, 3
P1	2, 0, 0	3, 2, 2	1, 2, 2
P2	3, 0, 2	6, 0, 2	6, 0, 0
P3	2, 1, 1	2, 2, 2	0, 1, 1
P4	0, 0, 2	4, 3, 9	4, 3, 1

Available Resources

3, 3, 2

Answer : Refer to the HTML file

11. Please run the following P0 to P4:

Safety Algorithm Simulation

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Next Step

Reset

Process	Allocation	Need
P0	0, 1, 0	7, 4, 3
P1	2, 0, 0	1, 2, 2
P2	3, 0, 2	6, 0, 9
P3	2, 1, 1	4, 1, 1

Simulated Available (Work)

3, 3, 2

Completion (Finish)

P0:F P1:F
P2:F P3:F

Answer : Refer to the HTML file

12. Please run the following P1:

Resource-Request Algorithm

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Next Step

Reset

Process	Allocation	Need
P1	2, 0, 0	1, 2, 2

P1 Request

1, 0, 2

System Available

3, 3, 2

Answer : Refer to the HTML file